

[clang-format] --dry-run crashes on C file containing embedded NUL byte, but works fine without --dry-run flag #

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ClosedBug#188631

EditsMember

clang-format crashes in --dry-run mode on an input file containing a NUL byte in a block comment.

Observed behavior:

- clang-format repro.c does not crash.
- clang-format --dry-run repro.c crashes with a segmentation fault.

In detail, when it has a NUL character in a comment (and likely also not in a comment), clang-format crashes with

PLEASE submit a bug report to <https://github.com/llvm/llvm-project/issues/> and include the crash backtrace in Stack dump:

0. Program arguments: /home/ubuntu/llvm-clang-1/llvm-install/usr/local/bin/clang-format --dry-run repro.c

#0 0x000055fe391fbaa8 llvm::sys::PrintStackTrace(llvm::raw_ostream&, int) (/home/ubuntu/llvm-clang-1/llvm-install

#1 0x000055fe391f8df5 llvm::sys::RunSignalHandlers() (/home/ubuntu/llvm-clang-1/llvm-install/usr/local/bin/clang

#2 0x000055fe391fcb86 SignalHandler(int, siginfo_t*, void*) Signals.cpp:0:0

#3 0x00007f53b2b31420 __restore_rt (/lib/x86_64-linux-gnu/libpthread.so.0+0x14420)

#4 0x000055fe391c9469 llvm::SourceMgr::GetMessage(llvm::SMLoc, llvm::SourceMgr::DiagKind, llvm::Twine const&, l

#5 0x000055fe391ad267 clang::format::emitReplacementWarnings(clang::tooling::Replacements const&, llvm::StringRe

#6 0x000055fe391aa709 clang::format::format(llvm::StringRef, bool) ClangFormat.cpp:0:0

#7 0x000055fe391a8003 main (/home/ubuntu/llvm-clang-1/llvm-install/usr/local/bin/clang-format+0x28003)

#8 0x00007f53b253f083 __libc_start_main /build/glibc-LcI20x/glibc-2.31/csu/../csu/libc-start.c:342:3

#9 0x000055fe391a4e2e _start (/home/ubuntu/llvm-clang-1/llvm-install/usr/local/bin/clang-format+0x24e2e)

Segmentation fault (core dumped)

It took me a bit to understand what the issue was with formatting the file, but when opening it with vim -b, I saw:

```
#if defined(AAA) && (BBB + 3) > 9 /*^@comment */ && !defined(CCC)
int f(void) { return 1; }
#elseif defined(DDD) || defined(EEE)
int f(void) { return 2; } /* trailing */
#else
int f(void) { return 3; }
```

It seems that it is a bug in some of the functionality of --dry-run that is masked or does not exist if using other clang-format flags. Which explains it. Yet when just removing this option:

```
/home/ubuntu/llvm-clang-1/llvm-install/usr/local/bin/clang-format repro.c
#if defined(AAA) && (BBB + 3) > 9 /*comment */ && !defined(CCC)
int f(void) { return 1;
}
#elseif defined(DDD) || defined(EEE)
int f(void) { return 2; } /* trailing */
#else
int f(void) { return 3; }
```

I also tried to fix the "[" to "{" in the first definition of f, it then ends okay (i.e. int f(void) { return 1; }).
I also tried to keep the "[" but removed the NUL char, then it works okay:

```
ubuntu@fuzzer-03:~/git/H-Fuzz/crashes_clang-format$ /home/ubuntu/llvm-clang-1/llvm-install/usr/local/bin/clang-format repro.c
del.c:2:24: warning: code should be clang-formatted [-Wclang-format-violations]
int f(void) { return 1; }
                ^
```

Technical data:

- clang-format version 23.0.0git (<https://github.com/llvm/llvm-project.git> 64e75b1)
- Target: x86_64-pc-linux-gnu
- OS: Ubuntu 20

Assignees

HazardyKnusperkeks

Labels

clang-formatcrash

Type

Bug

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

Code with agent mode

[clang-format] Don't crash on an input with a NUL char

llvm/llvm-project

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Participants

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llvmbot added clang-format last week

llvmbot last week

@llvm/issue-subscribers-clang-format)

Details

MemberAuthor

del.c

repro.c

I added the two files in the report above.

EugeneZelenko added crash last week

EugeneZelenko added the Bug issue type last week

HazardyKnusperkeks self-assigned this last week

HazardyKnusperkeks added a commit that references this issue last week

[clang-format] Don't crash on an input with a NUL char ... 8fb20e4

HazardyKnusperkeks mentioned this last week

[clang-format] Don't crash on an input with a NUL char #188631

HazardyKnusperkeks closed this as completed in #188631 last week

HazardyKnusperkeks added a commit that references this issue last week

[clang-format] Don't crash on an input with a NUL char (#188631) ... Verified 049700f

llvm-sync added a commit that references this issue last week

Automerger: [clang-format] Don't crash on an input with a NUL char (#1 ... 5167514

ambergorzynski added a commit that references this issue last week

[clang-format] Don't crash on an input with a NUL char (llvm#188631) ... b0350a3

Aadarsh-Keshri added a commit that references this issue 5 days ago

[clang-format] Don't crash on an input with a NUL char (llvm#188631) ... 11a7596

fzou1 added a commit that references this issue 4 days ago

[clang-format] Don't crash on an input with a NUL char (llvm#188631) ... ddb555a

Himadhith added a commit that references this issue 14 hours ago

[clang-format] Don't crash on an input with a NUL char (llvm#188631) ... f09b6be

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